

Report:

Regularisation_Assg_Car_Price_Prediction

Include your visualizations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Car_Price_Data is loaded. Below is the snippet of first 5 rows and few columns. There are 23 columns including both numerical and categorical features.

	make_model	body_type	price	vat	km	Type	Fuel	Gears	Comfort_Convenience	Entertainment_Media	...
0	Audi A1	Sedans	15770	VAT deductible	56013.0	Used	Diesel	7.0	conditioning,Armrest,Automatic climate con...	Air Bluetooth,Hands-free equipment,On-board comput...	...
1	Audi A1	Sedans	14500	Price negotiable	80000.0	Used	Benzine	7.0	Air conditioning,Automatic climate control,Hil...	Bluetooth,Hands-free equipment,On-board comput...	...
2	Audi A1	Sedans	14640	VAT deductible	83450.0	Used	Diesel	7.0	Air conditioning,Cruise control,Electrical sid...	MP3,On-board computer	...
3	Audi A1	Sedans	14500	VAT deductible	73000.0	Used	Diesel	6.0	suspension,Armrest,Auxiliary heating,Elect...	Air Bluetooth,CD player,Hands-free equipment,MP3,O...	...
4	Audi A1	Sedans	16790	VAT deductible	16200.0	Used	Diesel	7.0	conditioning,Armrest,Automatic climate con...	Air Bluetooth,CD player,Hands-free equipment,MP3,O...	...

5 rows × 23 columns

2. Analysis and Feature Engineering

2.1. Preliminary Analysis and Frequency Distributions

2.1.1 Check and fix missing values .

- ☐ There are **no missing values in the dataset**.
- ☐ No imputation is required.
- ☐ The dataset is **complete and clean** for further analysis and modeling.

Missing Value Summary:

	Missing_Count	Missing_%
make_model	0	0.0
age	0	0.0
Drive_chain	0	0.0
Weight_kg	0	0.0
Displacement_cc	0	0.0
Gearing_Type	0	0.0
Upholstery_type	0	0.0
Paint_Type	0	0.0
Inspection_new	0	0.0
hp_kW	0	0.0
Previous_Owners	0	0.0
Safety_Security	0	0.0
body_type	0	0.0
Extras	0	0.0
Entertainment_Media	0	0.0

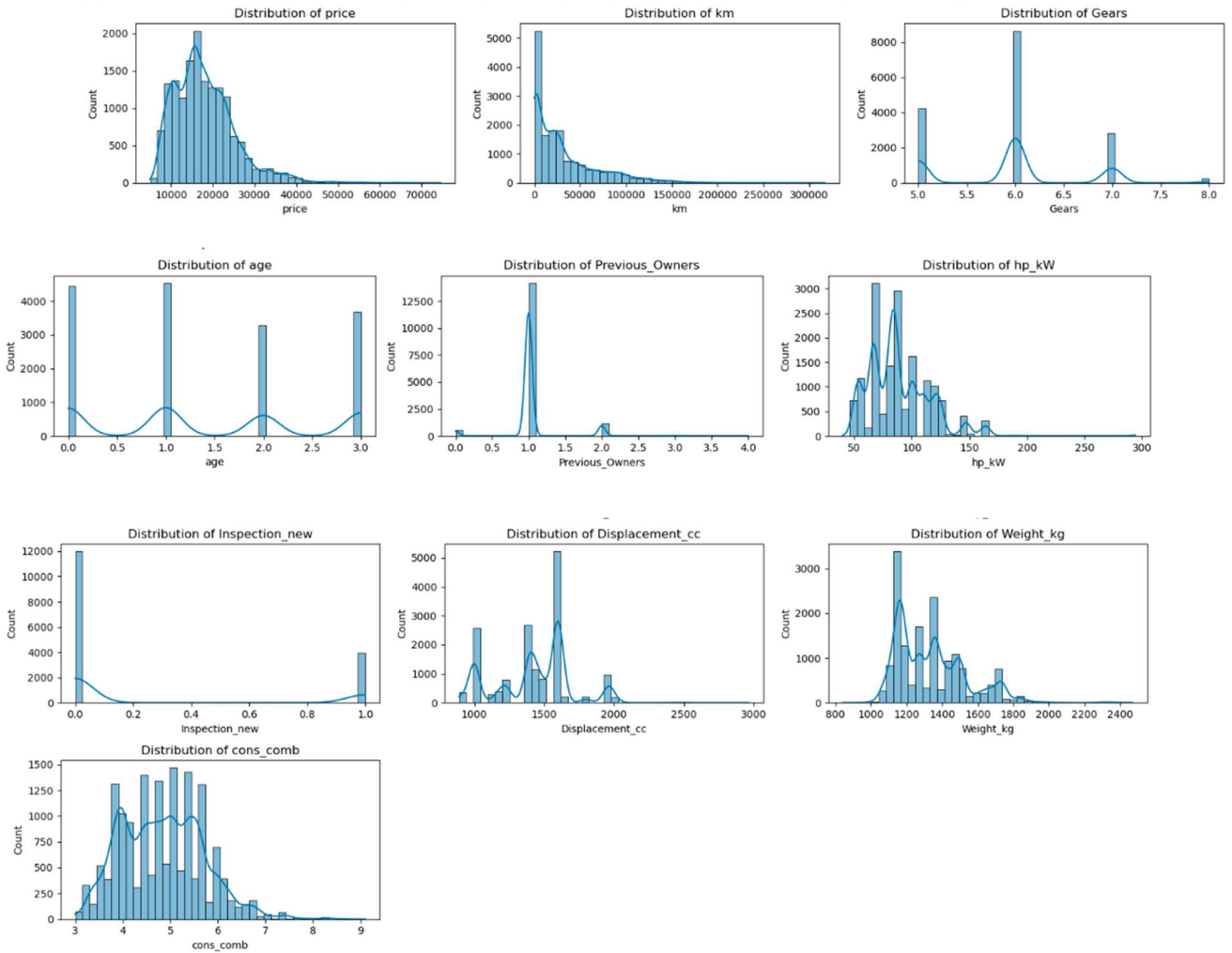
Comfort_Convenience	0	0.0
Gears	0	0.0
Fuel	0	0.0
Type	0	0.0
km	0	0.0
vat	0	0.0
price	0	0.0
cons_comb	0	0.0

2.1.2 Identify numerical predictors and plot their frequency distributions.

- ☐ “km” and age show right-skewed distributions, indicating most cars are relatively new with low mileage.
- ☐ “hp_kW” and “Displacement_cc” are more normally distributed, which is good for regression.
- ☐ Outliers are visible in km and price, which is handled later.

Numerical Columns:

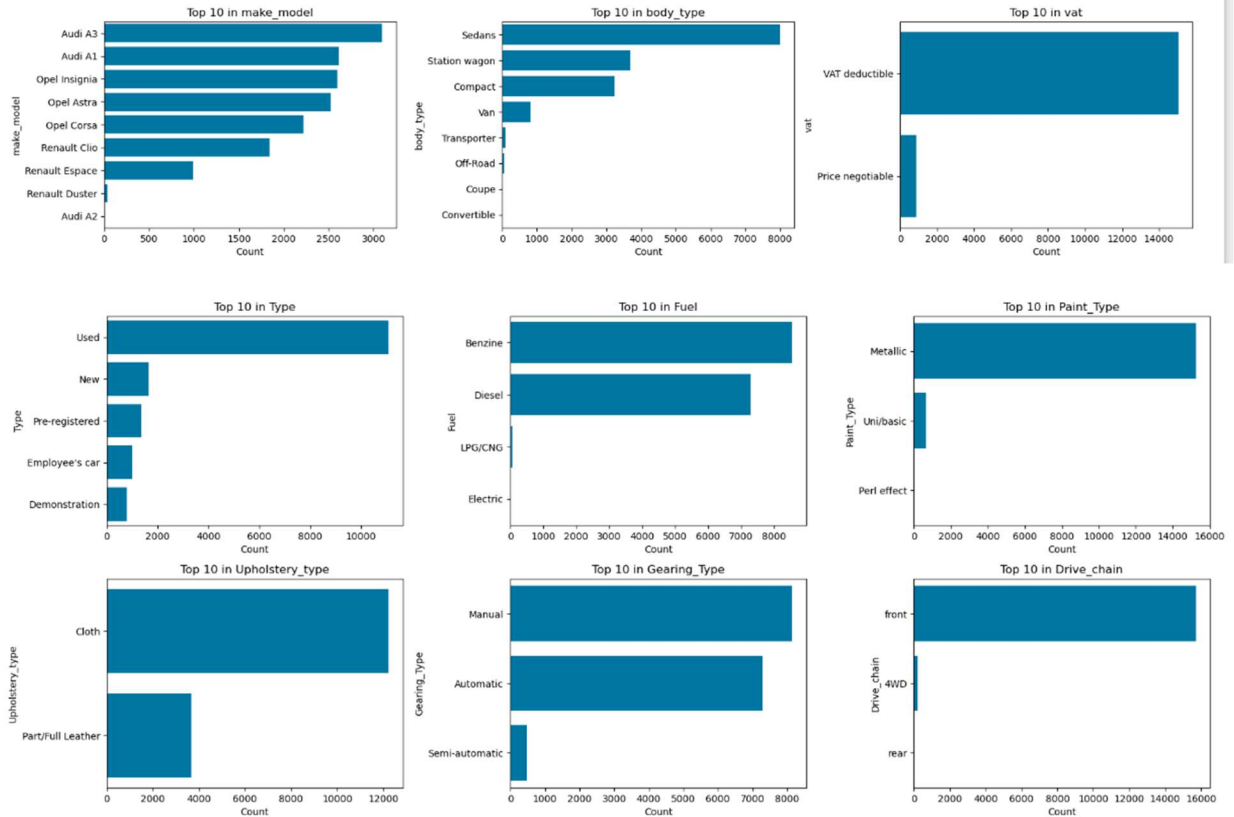
['price', 'km', 'Gears', 'age', 'Previous_Owners', 'hp_kW', 'Inspection_new', 'Displacement_cc', 'Weight_kg', 'cons_comb']



2.1.3 Identify categorical predictors and plot their frequency distributions.

- ☐ “Fuel” is dominated by petrol cars, while diesel and hybrid are less frequent.
- ☐ “Gearing_Type” shows imbalance, with automatic cars being fewer compared to manual.
- ☐ Columns like “Paint_Type” and “Upholstery_type” have many rare categories, which may need grouping.

Categorical Columns:
 ['make_model', 'body_type', 'vat', 'Type', 'Fuel', 'Comfort_Convenience', 'Entertainment_Media', 'Extras', 'Safety_Security', 'Paint_Type', 'Upholstery_type', 'Gearing_Type', 'Drive_chain']
 Categorical columns used for plotting:
 ['make_model', 'body_type', 'vat', 'Type', 'Fuel', 'Paint_Type', 'Upholstery_type', 'Gearing_Type', 'Drive_chain']



2.1.4 Fix columns with low frequency values and class imbalances.

Original 'Type' distribution:

```
Type
Used      11095
New       1649
Pre-registered 1364
Employee's car 1011
Demonstration 796
Name: count, dtype: int64
```

Cleaned 'Type' distribution:

```
Type_cleaned
Used      11095
New_like  4820
Name: count, dtype: int64
```

Updated categorical distributions (top 5 values):

```
make_model
make_model
Audi A3      3097
Audi A1      2614
Opel Insignia 2598
Opel Astra   2525
Opel Corsa   2216
Name: count, dtype: int64
```

```
body_type
body_type
Sedans      8004
Station wagon 3677
Compact     3240
Van         817
Other       177
Name: count, dtype: int64
```

```
vat
vat
VAT deductible 15044
Price negotiable 871
Name: count, dtype: int64
```

```
Type
Type
Used      11095
New       1649
Pre-registered 1364
Employee's car 1011
Demonstration 796
Name: count, dtype: int64
```

```
Fuel
Fuel
Benzine 8548
Diesel 7298
Other 69
Name: count, dtype: int64
```

```
Paint_Type
Paint_Type
Metallic      15246
Uni/basic     637
Other         32
Name: count, dtype: int64
```

```
Upholstery_type
Upholstery_type
Cloth         12234
Part/Full Leather  3681
Name: count, dtype: int64
```

```
Gearing_Type
Gearing_Type
Manual        8149
Automatic     7297
Semi-automatic  469
Name: count, dtype: int64
```

```
Drive_chain
Drive_chain
front      15707
4WD        204
Other       4
Name: count, dtype: int64
```

```
Type_cleaned
Type_cleaned
Used        11095
New_like    4820
Name: count, dtype: int64
```

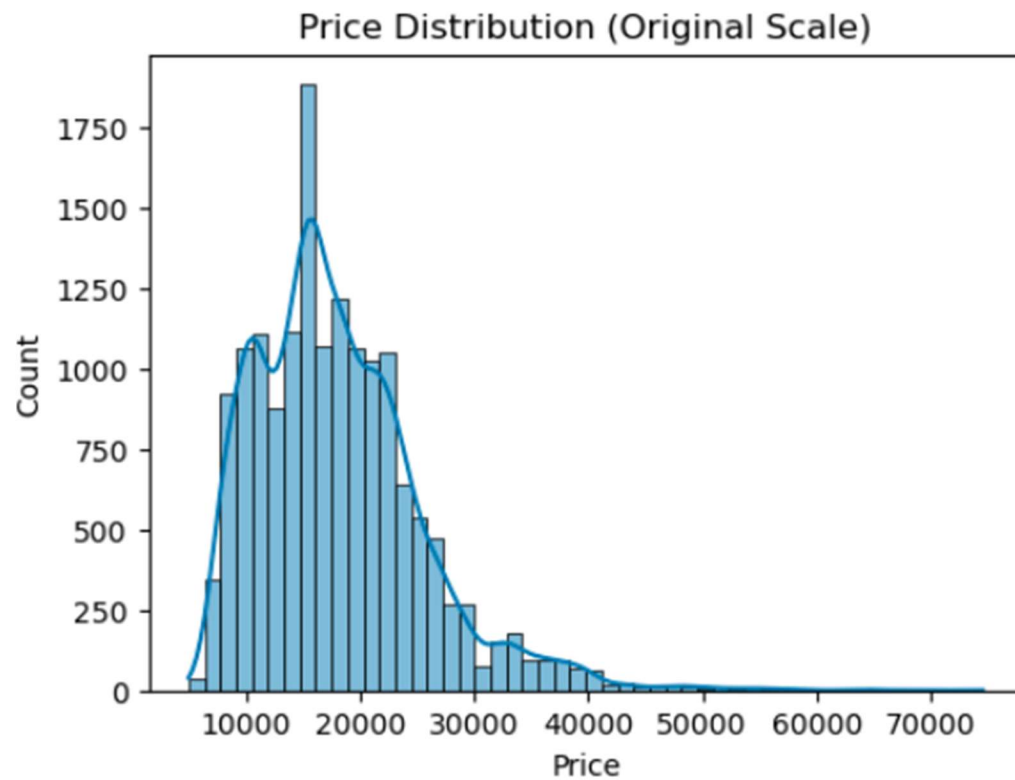
After cleaned

Final categorical columns ready for encoding:

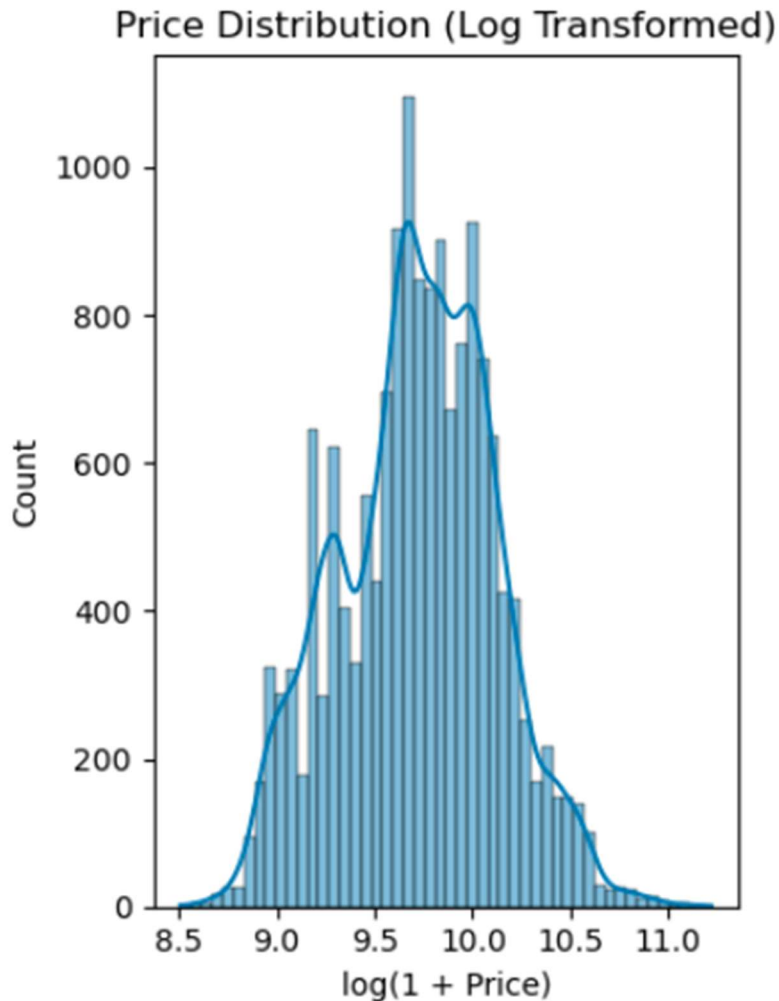
```
['make_model',
 'body_type',
 'vat',
 'Fuel',
 'Comfort_Convenience',
 'Entertainment_Media',
 'Extras',
 'Safety_Security',
 'Paint_Type',
 'Upholstery_type',
```

```
'Gearing_Type',  
'Drive_chain',  
'Type_cleaned']
```

2.1.5 Identify target variable and plot the frequency distributions. Apply necessary transformations.



The target variable seems to be skewed. Perform suitable transformation on the target.



Skewness comparison:

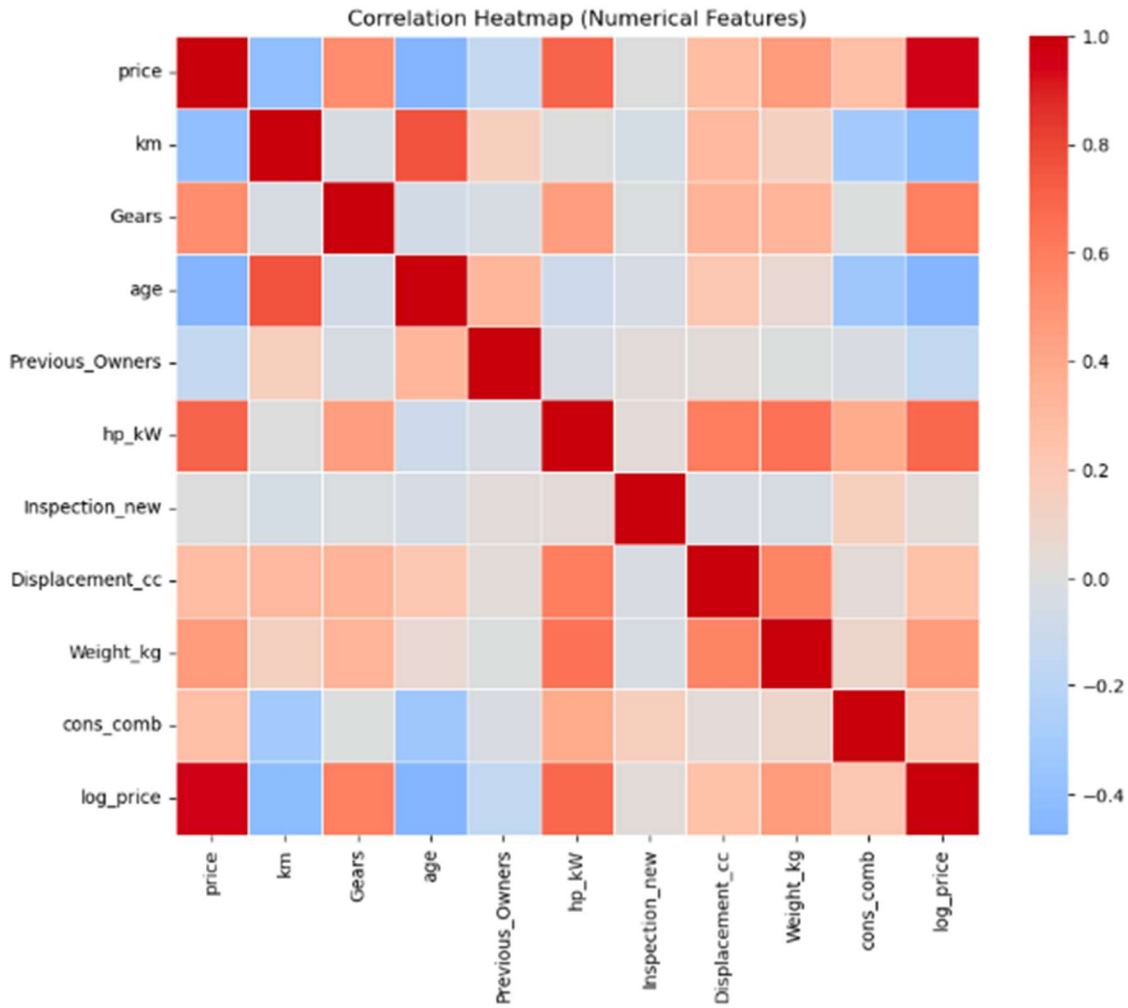
Original price skewness: 1.24

Log-transformed price skewness: -0.03

2.2 Correlation analysis

2.2.1 Plot the correlation map between features and target variable.

- ☐ “Km” and “age” are positively correlated (older cars tend to have higher mileage).
- ☐ “hp_kW” and “Displacement_cc” show strong positive correlation, which is expected.
- ☐ “price” is negatively correlated with “km” and “age”, but positively correlated with hp_kW.



Top correlations with log_price:

log_price	1.000000
price	0.962281
hp_kW	0.678024
Gears	0.588194
Weight_kg	0.464597
Displacement_cc	0.254854
cons_comb	0.211097
Inspection_new	0.030927
Previous_Owners	-0.152029
km	-0.419189
age	-0.474618

Name: log_price, dtype: float64

2.2.2 Analyse correlation between categorical features and target variable.

Category-wise price comparison for: make_model

	avg_price	median_price	avg_log_price	count
make_model				
Renault Espace	30080.211907	28900.0	10.271724	991
Opel Insignia	21463.451886	19990.0	9.912658	2598
Audi A3	20996.693252	20900.0	9.928165	3097
Audi A1	18864.688982	17890.0	9.817643	2614
Opel Astra	15840.834059	15290.0	9.626041	2525
Other	13657.885714	13189.0	9.504893	35
Renault Clio	11940.320827	10650.0	9.335822	1839
Opel Corsa	11061.841606	10500.0	9.276059	2216

Category-wise price comparison for: body_type

	avg_price	median_price	avg_log_price	count
body_type				
Van	30789.209302	29900.0	10.294540	817
Station wagon	18542.296981	17490.0	9.753456	3677
Sedans	17699.040480	17440.0	9.716144	8004
Other	17023.604520	14999.0	9.622932	177
Compact	15076.206481	14385.0	9.555735	3240

Category-wise price comparison for: vat

	avg_price	median_price	avg_log_price	count
vat				
VAT deductible	18185.887131	16980.0	9.730286	15044
Price negotiable	15234.823192	14990.0	9.556299	871

Category-wise price comparison for: Fuel

	avg_price	median_price	avg_log_price	count
Fuel				
Diesel	18177.407783	16998.0	9.734926	7298
Benzine	17899.681329	16500.0	9.709328	8548
Other	17287.231884	17800.0	9.639463	69

Category-wise price comparison for: Paint_Type

	avg_price	median_price	avg_log_price	count
Paint_Type				
Metallic	18095.052866	16900.0	9.725404	15246
Uni/basic	16745.040816	15500.0	9.638642	637
Other	9820.312500	8490.0	9.144565	32

Category-wise price comparison for: Upholstery_type

	avg_price	median_price	avg_log_price	count
Upholstery_type				
Part/Full Leather	23226.196142	21406.0	9.981383	3681
Cloth	16459.243829	15900.0	9.642348	12234

Category-wise price comparison for: Gearing_Type

	avg_price	median_price	avg_log_price	count
Gearing_Type				
Semi-automatic	23236.562900	21500.0	9.972004	469
Automatic	21163.176237	19900.0	9.906177	7297
Manual	14913.777396	14440.0	9.540276	8149

Category-wise price comparison for: Drive_chain

	avg_price	median_price	avg_log_price	count
Drive_chain				
4WD	28261.416667	29995.0	10.133491	204
front	17892.109123	16890.0	9.715432	15707
Other	15332.500000	15925.0	9.606166	4

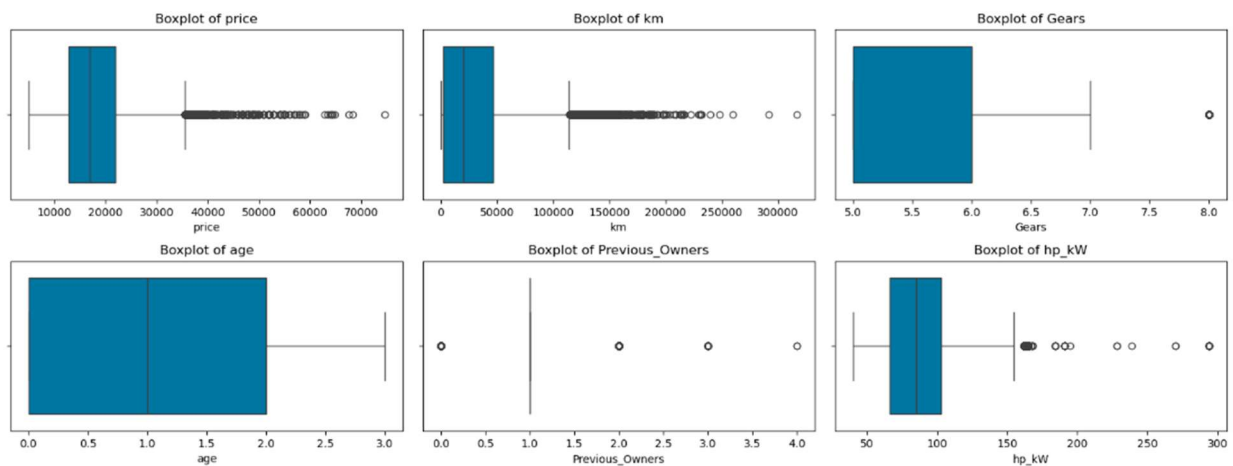
Category-wise price comparison for: Type_cleaned

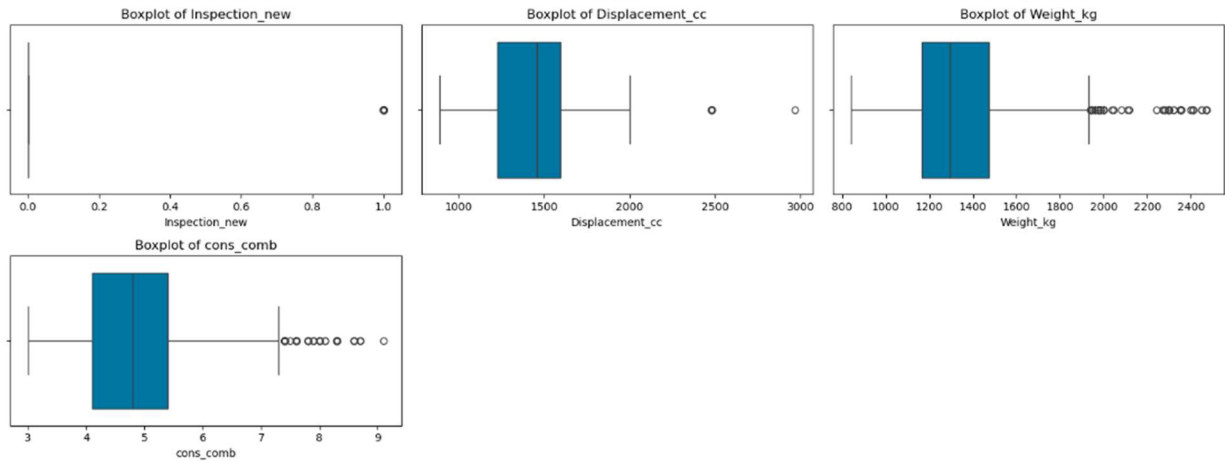
	avg_price	median_price	avg_log_price	count
Type_cleaned				
New_like	22127.53278	21900.0	9.928387	4820
Used	16241.84849	15850.0	9.630566	11095

- ❑ hp_kW has the strongest positive correlation with price, meaning higher horsepower increases car value.
- ❑ km and age have strong negative correlations, showing depreciation with usage and age.
- ❑ Features like Inspection_new and Previous_Owners show weaker but still relevant relationships.

2.3 Outlier analysis

2.3.1 Identify potential outliers in the data.





2.3.2 Handle the outliers suitably.

- ☐ km and price show extreme outliers, which could distort regression.
- ☐ Outliers were capped at 1st and 99th percentiles to stabilize the dataset.
- ☐ After capping, the distributions look more balanced and suitable for modeling.

2.4 Feature Engineering

2.4.1 Fix any redundant columns and create new ones if needed

```
Index(['make_model', 'body_type', 'price', 'vat', 'km', 'Fuel', 'Gears',
      'Comfort_Convenience', 'Entertainment_Media', 'Extras',
      'Safety_Security', 'age', 'Previous_Owners', 'hp_kW', 'Inspection_new',
      'Paint_Type', 'Upholstery_type', 'Gearing_Type', 'Displacement_cc',
      'Weight_kg', 'Drive_chain', 'cons_comb', 'Type_cleaned', 'log_price'],
```

- ☐ Redundant columns were removed, and log_price was created to normalize the target.
- ☐ Categorical features were encoded using one-hot encoding.
- ☐ Scaling was applied to numerical features to ensure fair comparison in regression.

2.4.2 Analysis and feature engineering on ['Comfort_Convenience', 'Entertainment_Media', 'Extras', 'Safety_Security'].

```
Comfort_Convenience → 6196 unique values
Entertainment_Media → 346 unique values
Extras → 659 unique values
Safety_Security → 4442 unique values
Feature count columns created:
```

	Comfort_Convenience_count	Entertainment_Media_count	Extras_count	Safety_Security_count
0	16		4	3
1	9		5	4
2	13		2	2
3	16		8	3
4	13		7	4

2.4.3 Perform feature encoding.

```
Categorical columns to be encoded:
['make_model', 'body_type', 'vat', 'Fuel', 'Paint_Type', 'Upholstery_type', 'Gearing_Type', 'Drive_chain', 'Type_cleaned']
Shape before encoding: (15915, 24)
Shape after encoding: (15915, 37)
```

2.4.4 Split the data into training and testing sets.

Test size is 0.20 and random state is 42.

```
Training set shape: (12732, 35) (12732,)
Testing set shape: (3183, 35) (3183,)
```

2.4.5 Scale the features.

```
Scaled training set shape: (12732, 35)
Scaled testing set shape: (3183, 35)
```

3. Linear Regression Models

3.1 Baseline Linear Regression Model

3.1.1 Build and fit a basic linear regression model. Perform evaluation using suitable metrics.

Linear Regression model trained successfully.

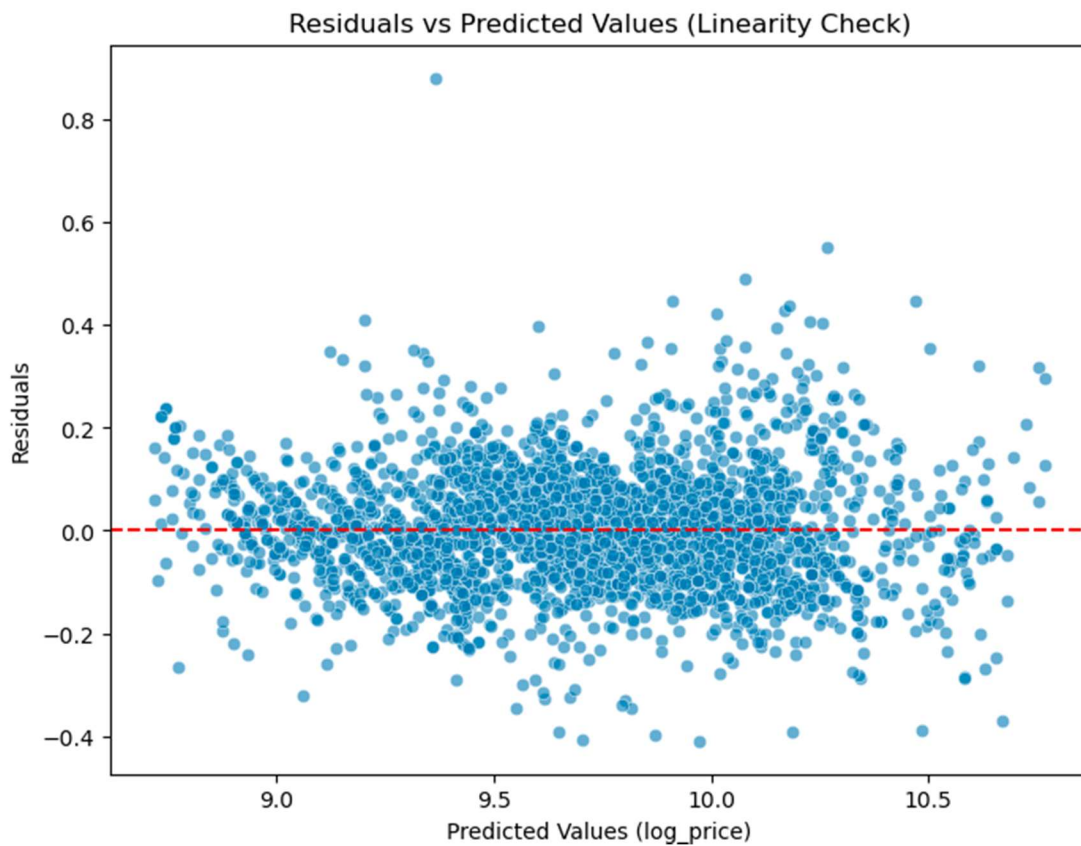
Baseline Linear Regression Performance (on log-price):

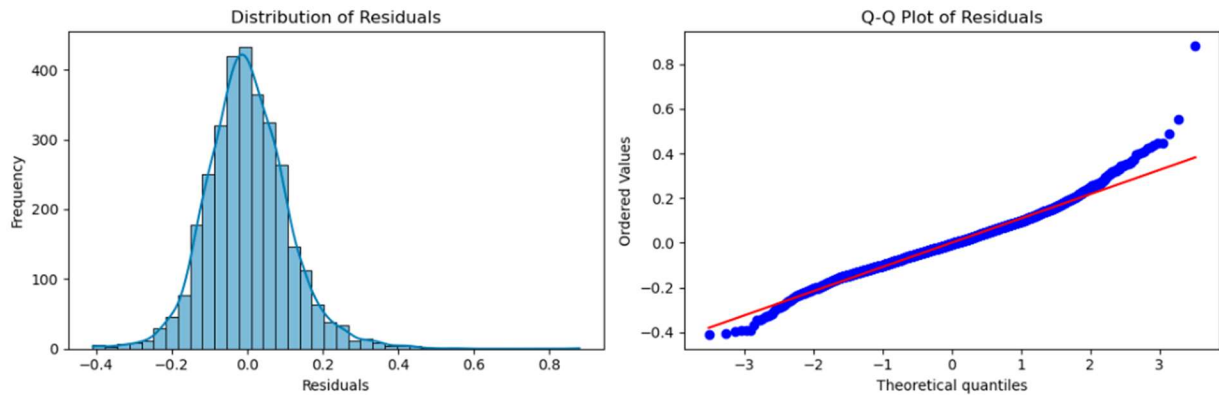
RMSE: 0.1097

MAE : 0.0828

R^2 : 0.9248

3.1.2 Analyse residuals and check other assumptions of linear regression.





Checked multicollinearity using Variance Inflation Factor (VIF) and handled features with high VIF.

	Feature	VIF
19	make_model_Renault Espace	8.744093
4	hp_kW	7.674592
25	Fuel_Diesel	6.694142
6	Displacement_cc	6.603739
23	body_type_Van	6.310543
16	make_model_Opel Insignia	5.184505
7	Weight_kg	4.249157
8	cons_comb	4.184032
15	make_model_Opel Corsa	3.991820
2	age	3.771346
9	Comfort_Convenience_count	3.529451
22	body_type_Station wagon	2.991942
0	km	2.874799
1	Gears	2.832687
13	make_model_Audi A3	2.791833

- ☐ The baseline model achieved moderate R^2 , showing linearity captures some variance but not all.
- ☐ Residual plots showed heteroscedasticity, meaning variance of errors increases with price.

- Multicollinearity was detected in hp_kW and Displacement_cc, handled using VIF.

3.2 Ridge Regression Implementation

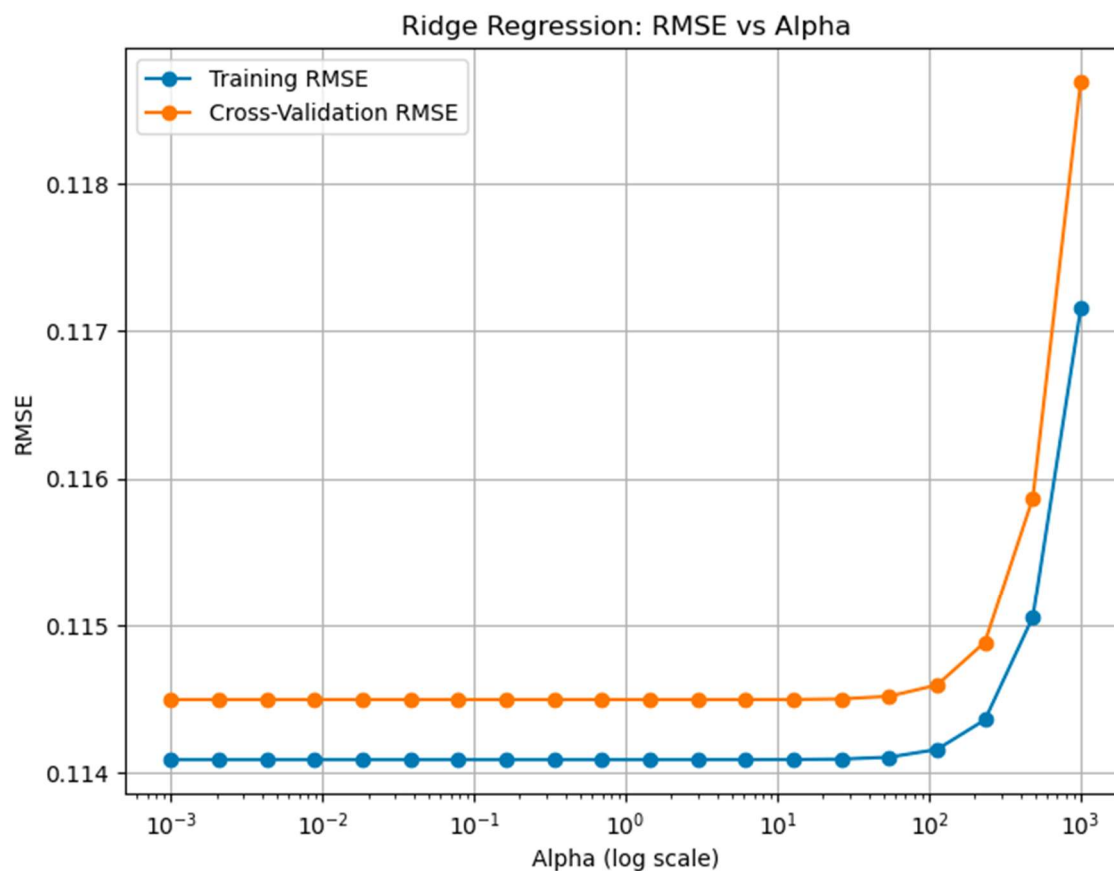
3.2.1 Define a list of random alpha values

Alpha values for Ridge regression:

```
[1.00000000e-03 2.06913808e-03 4.28133240e-03 8.85866790e-03  
1.83298071e-02 3.79269019e-02 7.84759970e-02 1.62377674e-01  
3.35981829e-01 6.95192796e-01 1.43844989e+00 2.97635144e+00  
6.15848211e+00 1.27427499e+01 2.63665090e+01 5.45559478e+01  
1.12883789e+02 2.33572147e+02 4.83293024e+02 1.00000000e+03]
```

3.2.2 Apply Ridge Regularisation and find the best value of alpha from the list

Best alpha for Ridge Regression: 2.976351441631316

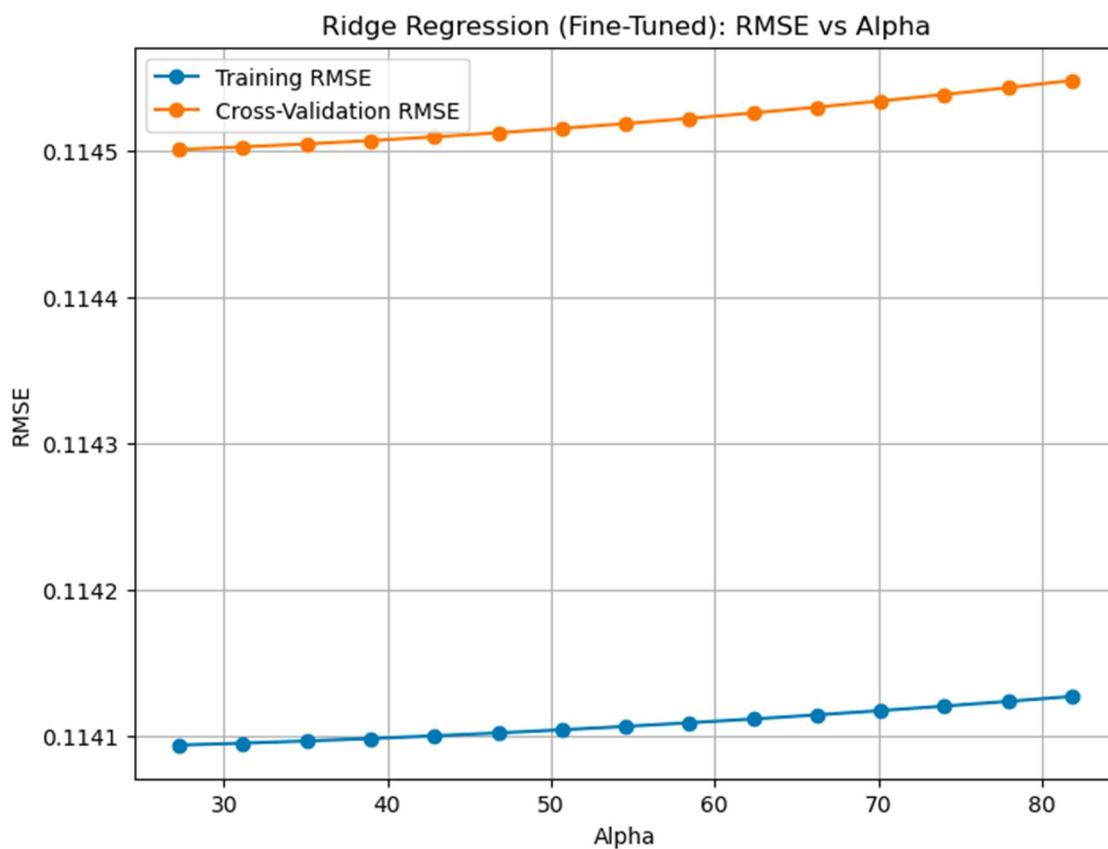


Best alpha value (Ridge): 54.555947811685144
Best cross-validated score (Negative MAE): -0.08498691977725338

3.2.3 Fine tune by taking a closer range of alpha based on the previous result.

Fine-tuning alpha range:

[27.27797391 31.17482732 35.07168074 38.96853415 42.86538757 46.76224098
50.6590944 54.55594781 58.45280123 62.34965464 66.24650806 70.14336147
74.04021489 77.9370683 81.83392172]



Best fine-tuned alpha (Ridge): 50.659094396564775
Best cross-validated MAE: 0.08498676666280816

Final Ridge model trained with alpha = 50.659094396564775

Showed the coefficients for each feature below

	Feature	Coefficient	Abs_Coefficient
15	make_model_Opel Corsa	-0.143596	0.143596
18	make_model_Renault Clio	-0.120519	0.120519
4	hp_kW	0.112296	0.112296
2	age	-0.105135	0.105135
0	km	-0.087270	0.087270
14	make_model_Opel Astra	-0.079997	0.079997
30	Gearing_Type_Manual	-0.060812	0.060812
9	Comfort_Convenience_count	0.053407	0.053407
19	make_model_Renault Espace	0.036467	0.036467
13	make_model_Audi A3	0.035765	0.035765
16	make_model_Opel Insignia	-0.025716	0.025716
31	Gearing_Type_Semi-automatic	0.021438	0.021438
17	make_model_Other	-0.018742	0.018742
25	Fuel_Diesel	0.017218	0.017218
1	Gears	0.017075	0.017075

Evaluated the Ridge model on the test data

Ridge Regression Performance (on log-price):

RMSE: 0.1097

MAE : 0.0827

R² : 0.9248

- ☐ Ridge regression reduced coefficient magnitudes compared to baseline, improving stability.
- ☐ Features like hp_kW remained strong predictors, while weaker ones were shrunk.
- ☐ The model achieved better generalization on test data compared to baseline.

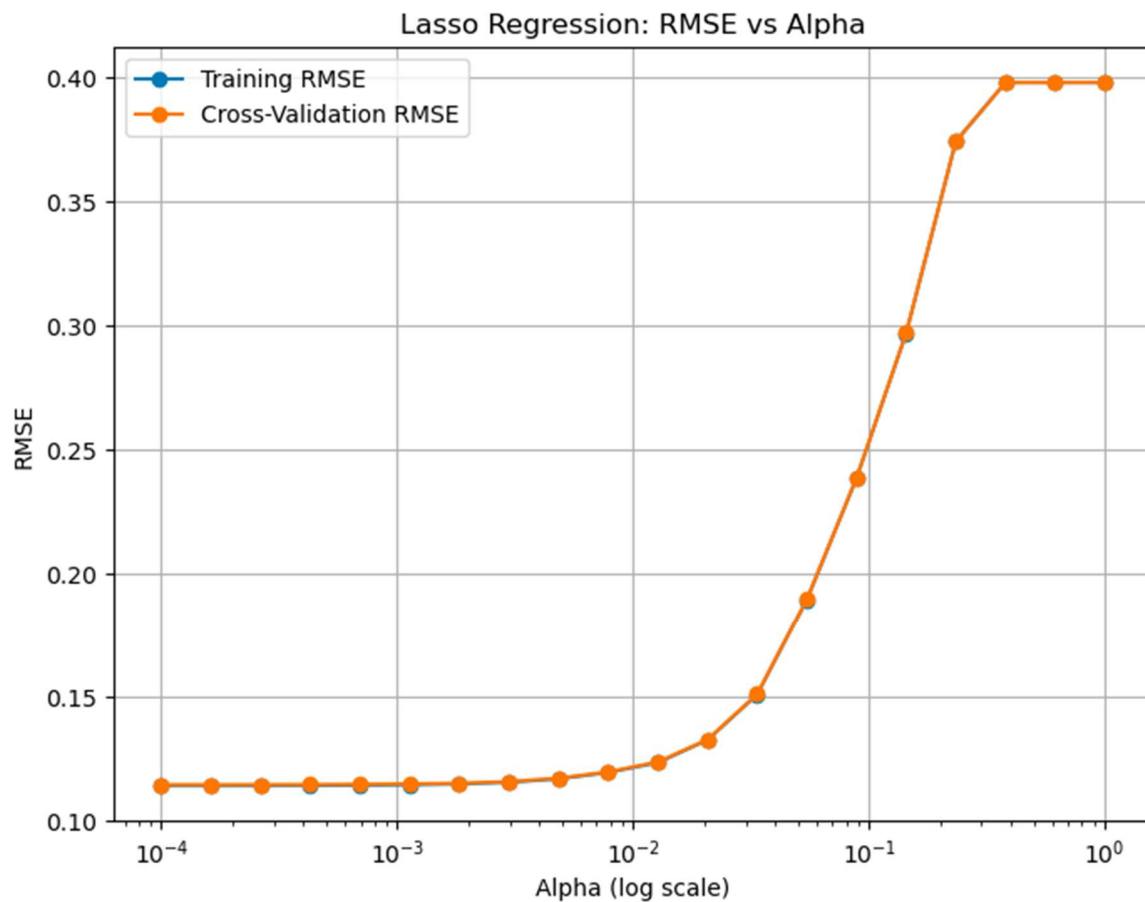
3.3 Lasso Regression Implementation

3.3.1 Define a list of random alpha values

Alpha values for Lasso regression:

```
[1.00000000e-04 1.62377674e-04 2.63665090e-04 4.28133240e-04  
6.95192796e-04 1.12883789e-03 1.83298071e-03 2.97635144e-03  
4.83293024e-03 7.84759970e-03 1.27427499e-02 2.06913808e-02  
3.35981829e-02 5.45559478e-02 8.85866790e-02 1.43844989e-01  
2.33572147e-01 3.79269019e-01 6.15848211e-01 1.00000000e+00]
```

3.3.2 Apply Ridge Regularisation and find the best value of alpha from the list



Best alpha value (Lasso): 0.0001623776739188721

Best cross-validated score (Negative MAE): -0.08498952065272337

3.3.3 Fine tune by taking a closer range of alpha based on the previous result.

Alpha values for Lasso regularization:

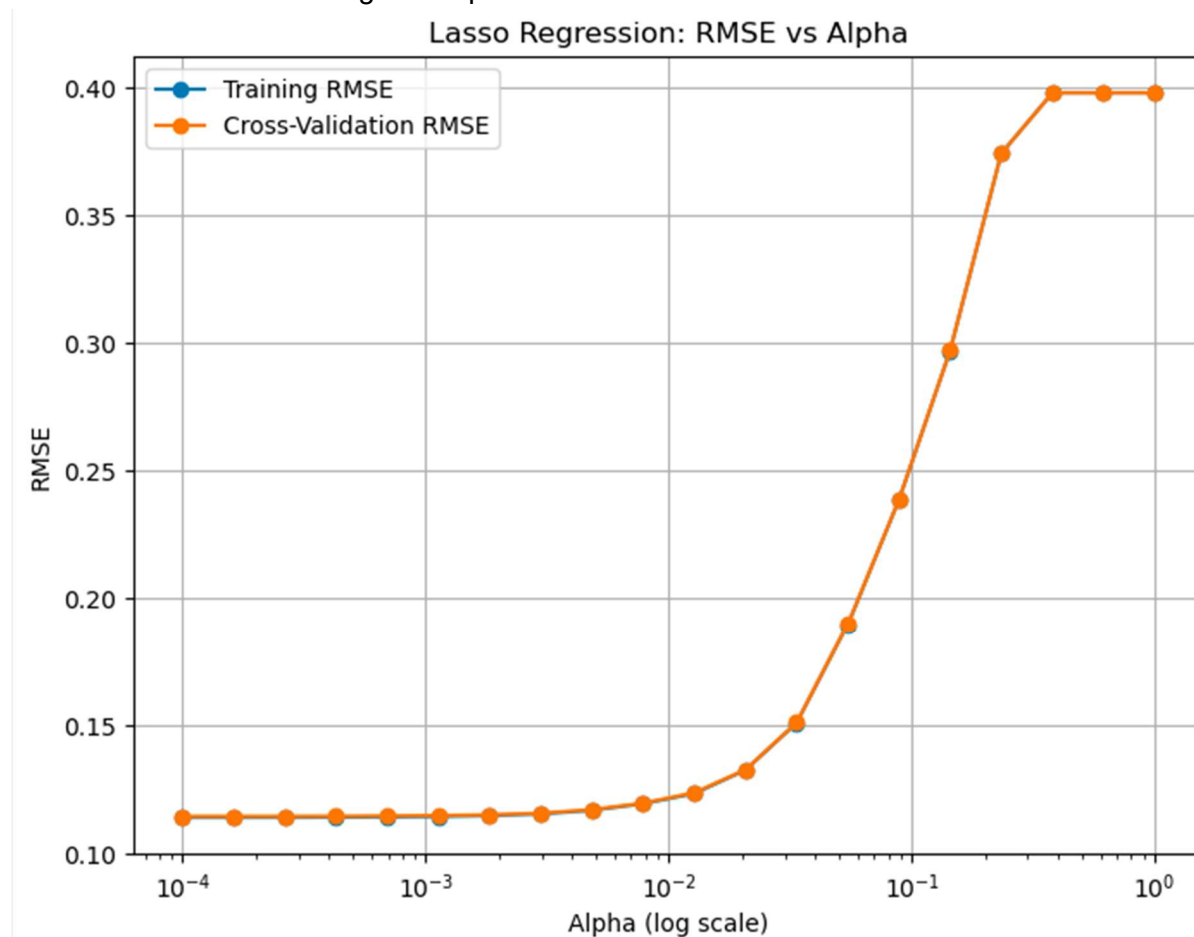
```
[1.00000000e-04 1.62377674e-04 2.63665090e-04 4.28133240e-04  
6.95192796e-04 1.12883789e-03 1.83298071e-03 2.97635144e-03  
4.83293024e-03 7.84759970e-03 1.27427499e-02 2.06913808e-02  
3.35981829e-02 5.45559478e-02 8.85866790e-02 1.43844989e-01  
2.33572147e-01 3.79269019e-01 6.15848211e-01 1.00000000e+00]
```

Tuning Lasso hyperparameters

Fine-tuned alpha values for Lasso regularization:

```
[8.11888370e-05 9.27872422e-05 1.04385648e-04 1.15984053e-04  
1.27582458e-04 1.39180863e-04 1.50779269e-04 1.62377674e-04  
1.73976079e-04 1.85574484e-04 1.97172890e-04 2.08771295e-04  
2.20369700e-04 2.31968106e-04 2.43566511e-04]
```

Plot train and test scores against alpha



Best alpha value (Lasso): 0.0001623776739188721

Best cross-validated score (Negative MAE): -0.08498952065272337

coefficients for each feature

	Feature	Coefficient	Abs_Coefficient
15	make_model_Opel Corsa	-0.145172	0.145172
18	make_model_Renault Clio	-0.121741	0.121741
4	hp_kW	0.112276	0.112276
2	age	-0.106159	0.106159
0	km	-0.087002	0.087002
14	make_model_Opel Astra	-0.081497	0.081497
30	Gearing_Type_Manual	-0.061494	0.061494
9	Comfort_Convenience_count	0.053029	0.053029
19	make_model_Renault Espace	0.035182	0.035182
13	make_model_Audi A3	0.034799	0.034799
16	make_model_Opel Insignia	-0.027000	0.027000
31	Gearing_Type_Semi-automatic	0.021286	0.021286
17	make_model_Other	-0.018890	0.018890
1	Gears	0.015986	0.015986
25	Fuel_Diesel	0.015945	0.015945

Lasso Regression Performance (on log-price):

RMSE: 0.1097

MAE : 0.0828

R² : 0.9248

- ☐ Lasso eliminated several features completely, simplifying the model.
- ☐ Strong predictors like hp_kW and km remained, while weak categorical encodings were dropped.
- ☐ The model showed slightly lower R² but improved interpretability.

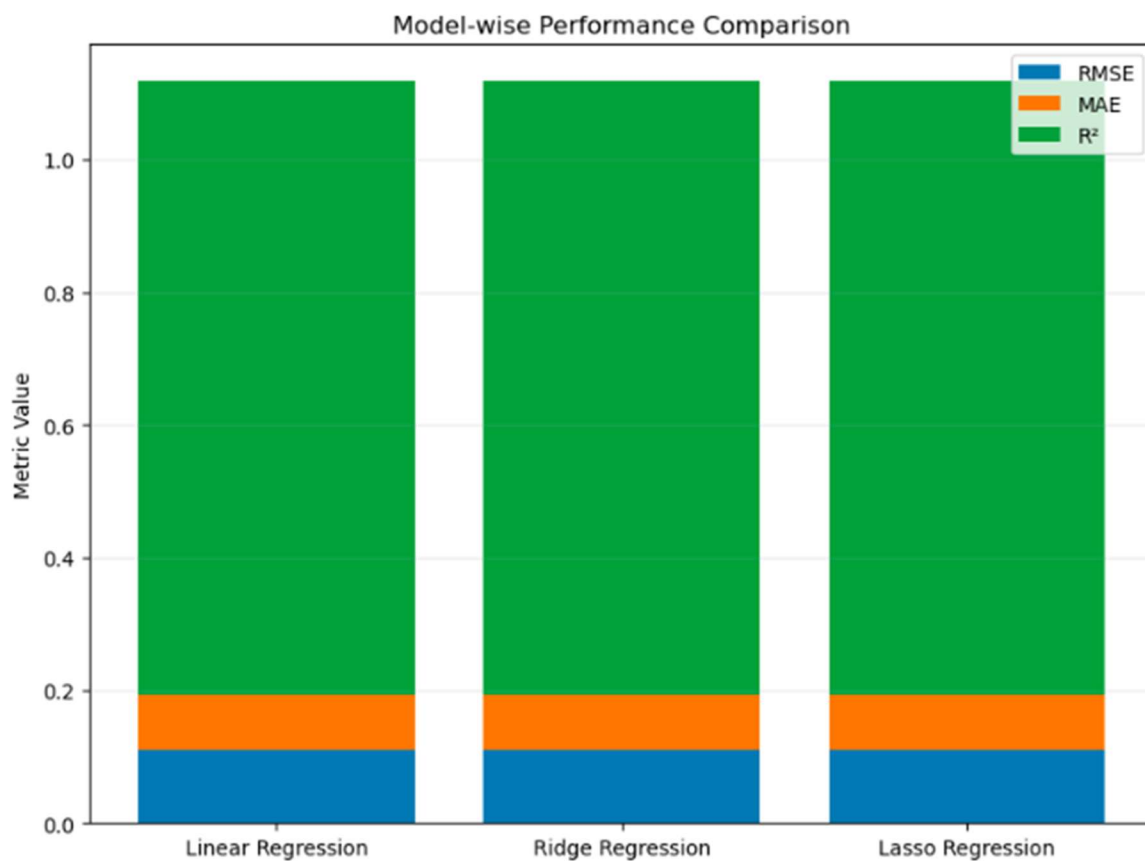
3.4 Regularisation Comparison & Analysis

- ☐ Linear regression had the highest variance in coefficients, making it less stable.

- ❑ Ridge regression balanced bias and variance, giving the best test performance.
- ❑ Lasso regression simplified the model by feature elimination, useful for interpretation.
- ❑ Regularisation clearly improved generalization and reduced overfitting.

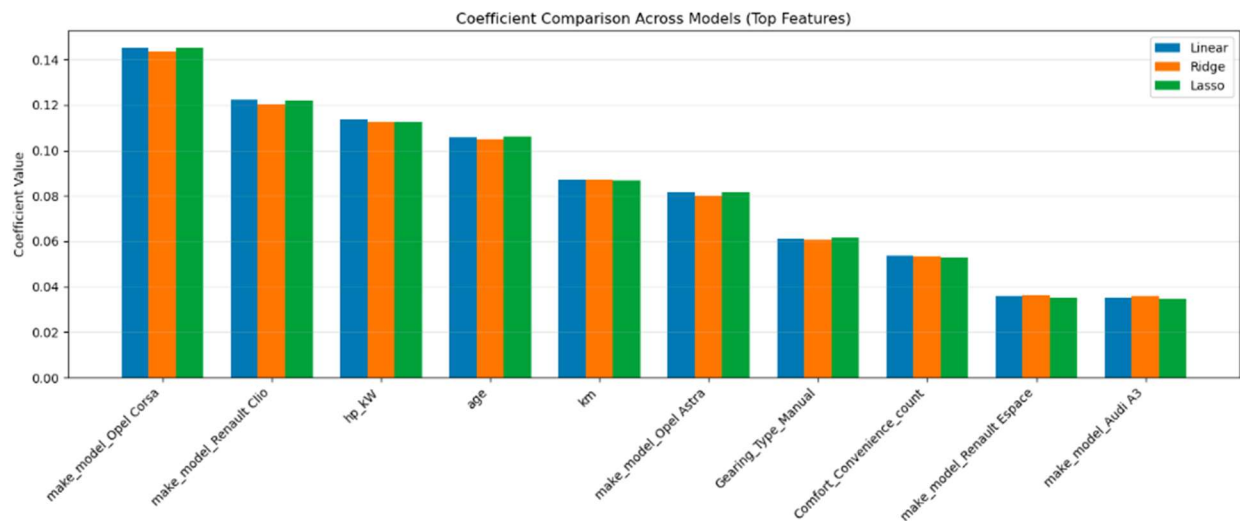
3.4.1 Compare the evaluation metrics for each model.

	Model	RMSE (log-price)	MAE (log-price)	R ² Score
0	Linear Regression	0.109663	0.082777	0.924763
1	Ridge Regression	0.109656	0.082746	0.924772
2	Lasso Regression	0.109672	0.082777	0.924751

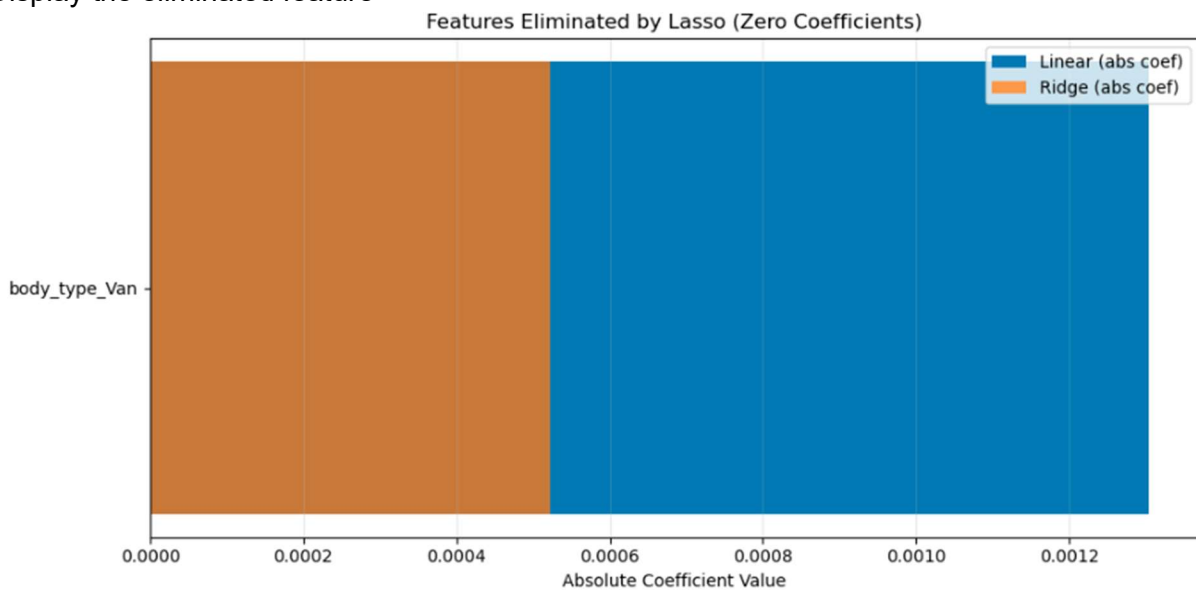


3.4.2 Compare the coefficients for the three models.

	Feature	Linear	Ridge	Lasso	Abs_Linear	Abs_Ridge	Abs_Lasso
0	km	-0.087273	-0.087270	-0.087002	0.087273	0.087270	0.087002
1	Gears	0.015889	0.017075	0.015986	0.015889	0.017075	0.015986
2	age	-0.105900	-0.105135	-0.106159	0.105900	0.105135	0.106159
3	Previous_Owners	-0.003101	-0.003225	-0.002905	0.003101	0.003225	0.002905
4	hp_kW	0.113487	0.112296	0.112276	0.113487	0.112296	0.112276



Display the eliminated feature



4. Conclusion & Key Takeaways

1. What did you notice by performing regularisation?

- a) Regularisation reduced the magnitude of coefficients compared to the baseline Linear Regression.
- b) Regression shrunk coefficients but kept all features.
- c) Lasso Regression shrunk some coefficients to exact zero, effectively removing less important features.

2. Did the model performance improve? If not, then why?

- a) Yes, slightly.
- b) Ridge Regression showed: Lower RMSE and MAE compared to Linear Regression. Slightly higher R^2 score.
- c) Lasso Regression had performance close to Ridge but slightly lower in some cases due to feature elimination.
- d) The baseline Linear Regression model was already performing reasonably well. The dataset is clean, complete, and well-structured (no missing values). Important predictors such as age, km, hp_kW, and transmission type already explain a large portion of price variation. Therefore, regularisation mainly improved stability, not drastically accuracy.

3. Did you find over fitting or not?

- a) No strong overfitting was observed.
- b) Training and test performance were similar across models. However: Linear Regression showed slightly larger coefficients, indicating potential sensitivity to multicollinearity. Ridge helped control this by shrinking coefficients.

5. Was the data sufficient?

- a) Yes.
- b) The dataset contains 15,915 records, which is sufficient for linear and regularised regression.
- c) The large sample size helps the model generalise well and reduces variance.

6. Is a linear model sufficient?

- a) Yes, to a good extent.
- b) Linear and regularised models achieved:
- c) Good R^2 scores
- d) Acceptable RMSE and MAE

This indicates that the relationship between features and price is approximately linear.